Pessimism and Business Cycles

Guangyu Pei, Yikai Wang

CUHK, Essex

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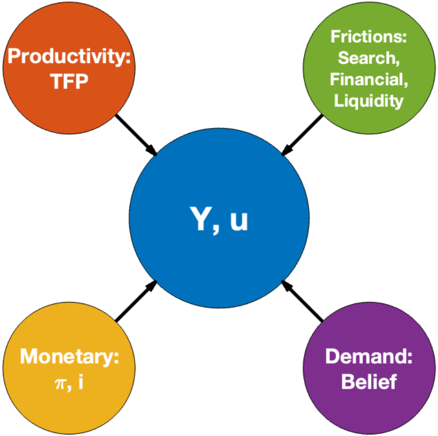
Business Cycles: Different Shocks



Main Business Cycle (MBC)

- Angeletos, Collard and Dellas (2019)
- PCA + VAR of u, Y, h, I, C; $Y/h, TFP; \pi, r$ ...
- First principle component (MBC), correlation
  - High (75~90%): u, Y, h, I
  - Little ($<$10%): π (real business cycle?)

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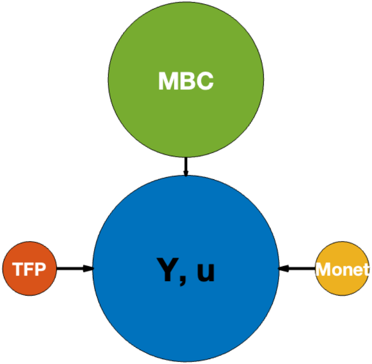
- First principle component (MBC), correlation
  - High (75~90%): u, Y, h, I
  - Little (<10%): π (real business cycle?)
  - Lower (<20%): TFP

Main Business Cycle: ACD (2019)'s View



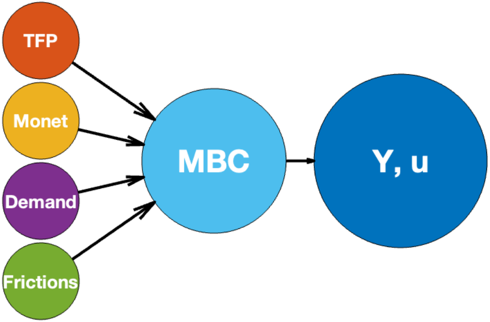
Main Business Cycle: ACD (2019)'s View

- "All business cycles are alike." - Lucas (1977)
- Low correlation with $TFP, \pi \Rightarrow$ **Demand, noninflationary**
- One large exogenous demand shock, e.g., sentiments
- Alternative: Different shocks, but the same propagation mechanism in demand side
  (Hard to imagine!)

"Thought the harder, heart the keener."

MBC: Alternative Story

- A common propagation mechanism
  - Any negative (real) shock $\Rightarrow$ **pessimism** $\Rightarrow$ demand drops
- **Ambiguity aversion** $\Rightarrow$ **pessimism** $\Rightarrow$ business cycles
  - Klibanoff, Marinacci and Mukerji (2005)
  - Pei (2021), Huo, Pei, Pedroni (2024)
- Alternative mechanisms to generate pessimism?
  - Behavioral expectation bias of poor?


TFP
Monet
Demand
Frictions
MBC
Y, u

What We Do

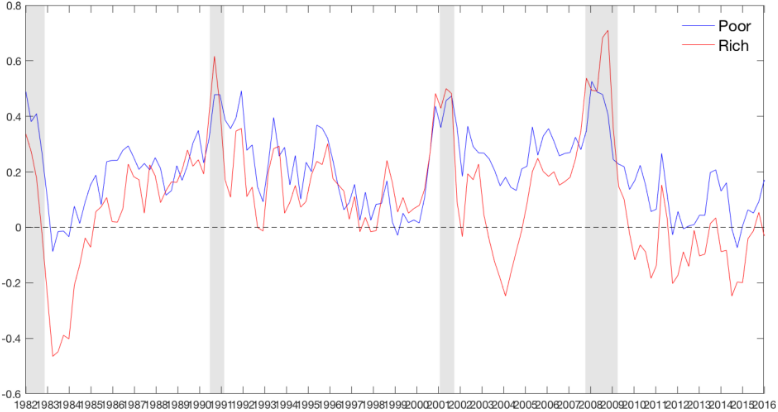
- Document empirical facts
  - Pessimism: in recession, for rich and poor households
  - PCA: pessimism as the MBC
- Build a theory of unified business cycles
  - with **endogenous** demand fluctuation
  - driven by **different exogenous (real) shocks**
  - consistent with business cycle comovement (Y/h, TFP, π)

Our Contribution

- Bridge RBC and demand driven business cycles
- Real shocks and demand fluctuation, united

	Real shocks	Demand Shocks
Pros	Interpretation/Policy	Data Fit
Cons	Y/H, TFP	Microfoundation/Interpretation

Pessimism



Poor
Rich
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016

Pessimism and Business Cycles

target	poor	mid	rich	u	Y	H	I	C	π
u	**0.15**	0.33	**0.46**	0.85	**0.73**	0.78	0.76	0.70	**0.11**
Y	**0.23**	0.36	**0.51**	**0.68**	0.92	0.69	0.83	0.93	**0.15**
H	0.16	0.30	0.40	0.75	0.72	0.88	0.75	0.68	0.11
I	0.15	0.32	0.47	0.74	0.87	0.75	0.87	0.85	0.13
C	0.23	0.35	0.49	0.64	0.91	0.65	0.79	0.94	0.16
π	0.13	0.14	0.10	0.06	0.07	0.02	0.07	0.06	0.61

Pessimism: Facts

1. Correlate with **output** over time (negative/counter-cyclical)
   - $Y \downarrow$ (recession), $Pess \uparrow$
2. Correlate with **household incomes** (negative)
   - $y^h \downarrow$, $Pess \uparrow$
3. **Gaps** between rich and poor are smaller during recessions
   - $Y \downarrow$, $\triangle Pess \downarrow$
4. Business cycle variables are more correlated with **pessimism of rich** than poor
   - $corr(Pess^r, Y) > corr(Pess^p, Y)$

Model: Households in a Partial Equilibrium

- Income process for household h:

$$y_t^h \equiv \log\left(Y_t^h\right) = \rho_y y_{t-1}^h + \lambda^h \zeta_t, \qquad \zeta_t \sim N\left(\mu_t, \sigma_\zeta^2\right),$$

  - y_t^h: incomes, labor + risky investments (e.g., stocks)
  - λ^h : load aggregate shocks, differ across households

- Ambiguity:

$$\mu_t \sim N\left(0, \sigma_\mu^2\right)$$

- Budget constraint

$$C_t + S_{t+1} = Y_t + (1+r)S_t$$

- Rich: λ^r; unconstrained

- Poor: λ^p, binding constraint:

$$S_{t+1}^p \geq \underline{S}^p.$$

Model

- Ambiguity aversion as a concave transformation of expected utility conditional on μ

$$\phi\left(\mathbb{E}^{\mu} V\right) = -\frac{1}{\lambda}\exp\left(-\lambda \mathbb{E}^{\mu} V\right)$$

- Household's problem at the beginning of period t

$$\begin{aligned} V\left(Y_t, S_t\right) = \max_{C_t} \quad & \frac{C_t^{1-\sigma}}{1-\sigma} \\ & + \beta \phi^{-1}\left(\int_{\mathbb{R}} \phi\left(\mathbb{E}_t^{\mu_{t+1}}\left[V\left(S_{t+1}, Y_{t+1}\right)\right]\right) f\left(\mu_{t+1}\right) d\mu_{t+1}\right) \end{aligned}$$

$$\text{s.t.} \quad C_t + S_{t+1} = Y_t + (1+r) S_t, \qquad S_{t+1}^p \geq \underline{S}^p.$$

Belief and Consumption

- Time consistent
  $\Rightarrow$ use distorted posterior belief

$$\widetilde{f}_t\left(\mu_{t+1}\right) \propto \exp\left(-\lambda \mathbb{E}_t^{\mu_{t+1}}\left[V\left(S_{t+1}, Y_{t+1}\right)\right]\right) f\left(\mu_{t+1}\right)$$

  instead of Bayesian belief $f\left(\mu_{t+1}\right)$

- First order condition:

$$C_t^{-\sigma} = \int_{\mathbb{R}} \mathbb{E}_t^{\mu_{t+1}}\left[C_{t+1}^{-\sigma}\right] \widetilde{f}_t\left(\mu_{t+1}\right) d\mu_{t+1}$$

- Consumption demand

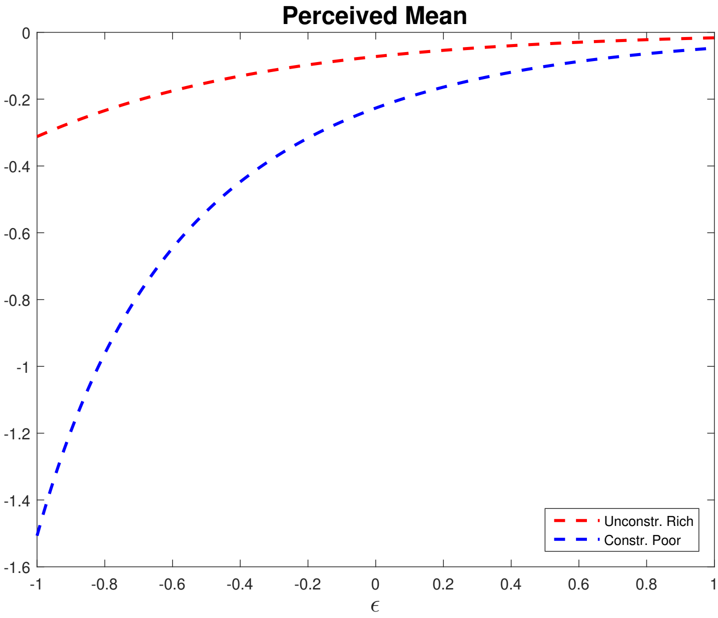
$$\widehat{c}_t = \int_{\mathbb{R}} \mathbb{E}_t^{\mu_{t+1}}\left[\widehat{c}_{t+1}\right] \widetilde{f}_t\left(\mu_{t+1}\right) d\mu_{t+1}$$

Pessimism

$\widetilde{f}_t(\mu_{t+1})$: negatively related to $\mathbb{E}_t^{\mu_{t+1}}[V(S_{t+1}, Y_{t+1})]$

1. Households are pessimistic
   - $\mu_{t+1}\downarrow$, $\widetilde{f}_t(\mu_{t+1})\uparrow$
   - $\widehat{c}_t < c_t$
2. More pessimistic with negative shocks
   - $Y\downarrow, \widetilde{\mathbb{E}}\mu_{t+1}\downarrow$
3. Rich are generally less pessimistic than poor
   - $\widetilde{\mathbb{E}}^r\mu_{t+1} > \widetilde{\mathbb{E}}^p\mu_{t+1}$ if $\lambda^r = \lambda^p$

Shocks: $\lambda^r = \lambda^p$



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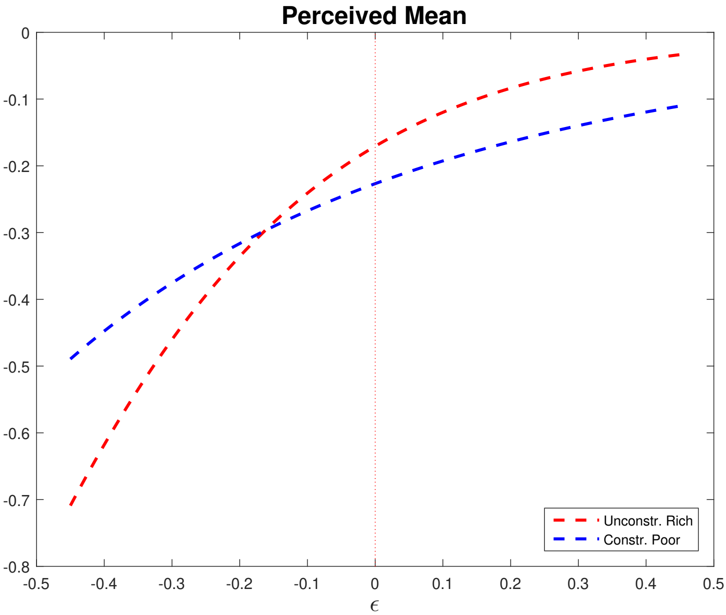

Pessimism Gaps: Rich & Poor

- Rich can become as pessimistic as poor in recessions
  - if recessions is associated with larger shocks with the rich
- Large shocks to rich are necessary for recessions
  - Large shocks to rich $\Rightarrow$ Recessions
  - Recessions $\Rightarrow$ Larger shocks to rich than poor

General Equilibrium: Demand ⇒ Employment

- Angeletos and La'O (2013); Pei (2021)
  - Monopolistic competition between islands j
  - firms and workers observe local productivity a_j
  - decides local labor demand and supply
  - after local production finishes, trade and find prices that clear markets
- Or HANK/TANK?

Conclusion

- Pessimism
  - Correlate with BC
  - Differ across rich and poor: level and importance for BC
- Theory
  - Bridges real shocks and demand fluctuation
  - Non-inflationary
- Future work
  - General equilibrium